Mutual-Feed Learning for Super-Resolution and Object Detection in Degraded Aerial Imagery

Jinze Yang, Kun Fu, Youming Wu, Wenhui Diao, *Member, IEEE*, Wei Dai, and Xian Sun, *Senior Member, IEEE*

***Abstract*— The resolution degradation poses a huge challenge for object detection (OD) in aerial imagery. The existing methods use super-resolution (SR) based on generative adversarial network (GAN) to restore texture details in degraded images. However, constrained detection results are still acquired due to object feature difference between restored and clear images. Therefore, we propose a simple yet effective learning method called mutual-feed learning (MFL) to solve the problem in this article. A closed-loop structure is designed via building the feedback connection based on the feedforward connection between the two tasks. It effectively delivers the object spatial and feature information from OD to SR and provides restoration-enhanced images from SR to OD. Specifically, a feedback of region of interest (FROI) module is introduced to realize a region-level discrimination under the guidance of object information. It guides the discrimination process of SR. Furthermore, a multiscale object information (MSOI) module is developed to implement a feature-level restoration by narrowing the differences in object-related features. It improves the generation process of SR. Then OD can be performed in restoration-enhanced images to obtain more accurate results. Extensive experiments over Northwestern Polytechnical University (NWPU) Very High Resolution (VHR)-10, Cars Overhead With Context (COWC), and Fine-grained Object Recognition (FAIR)1M datasets show that the method can achieve state-of-the-art results.**

***Index Terms*— Closed-loop framework, feedback of region of interest (FROI), multiscale object information (MSOI), mutual-feed learning (MFL), object detection (OD), super-resolution (SR).**

I. Introduction

OBJECT detection is a fundamental task in the field of remote sensing. It can be applied to various applications: object tracking, change detection, etc. Satisfactory detection performance always relies on clear images (CI) of good quality. However, due to harsh imaging conditions—the aerial camera shake, the short transmission bandwidth, the long-range shooting, and the undersampled imaging in aerial imagery—degraded images [1], [2], [3], [4], [5] are often captured. Specifically, resolution degradation is a common type of degradation and has been widely concerned. As shown in Fig. 1, texture features are often missing in resolution degraded images. Object regions in degraded images are more blurred than those in CIs. Poor detection results are acquired in degraded images due to missing object features.

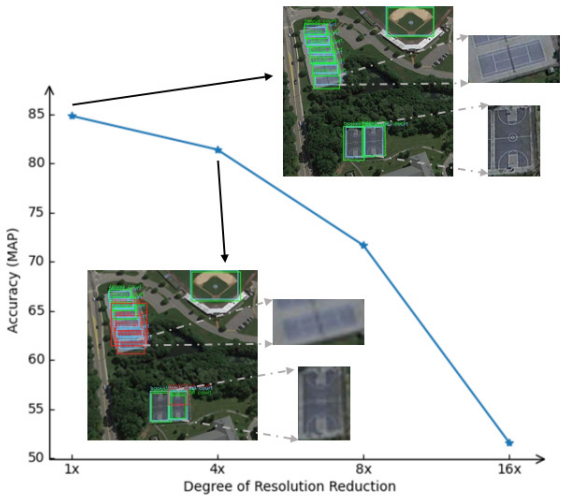


Fig. 1. Detection results in degraded images; 1x, 4x, 8x, and 16x represent the degrees of resolution degradation in degraded images compared with CIs; Cyan boxes show ground truth; red boxes show wrong detection; green boxes show correct detection.

To solve the problem of resolution degradation, super-resolution (SR) has been used to restore missing texture details in degraded images before object detection (OD). However, the existing methods based on generative adversarial network (GAN) still have some deficiencies. Rabbi *et al.* [6] propose to fine-tune parameters of the SR network through backpropagation of detection loss. However, the two tasks are optimized separately, and the fine-tuning operation may be affected by optimization of SR itself. Mostofa *et al.* [7] develop a multitask loss to optimize the two tasks together. It only applies detection loss as the low-level guidance for optimizing SR. It is not adequate for providing high-level guidance, such as the object information guidance which is significant for detection-driven SR. Bai *et al.* [8] propose to obtain object patches first. Then SR and OD are performed in these patches. Nevertheless, it is time-consuming and difficult to obtain object patches accurately. Therefore, in these existing methods, the optimization of generation and discrimination is not guided by object information. The detection-interested

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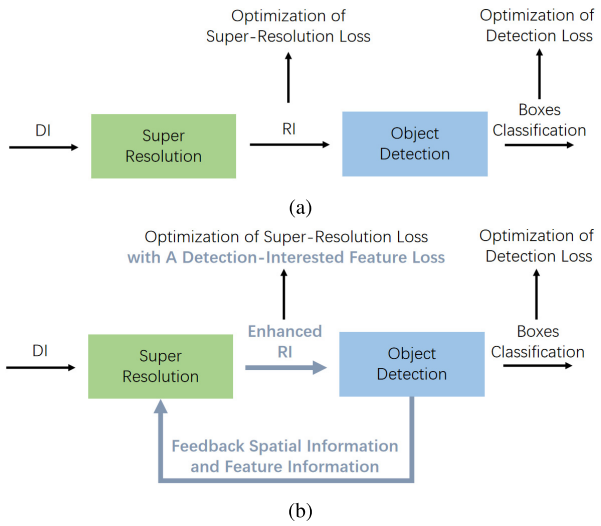


Fig. 2. (a) Framework of previous methods based on an open-loop structure. (b) Framework of MFL based on a closed-loop structure. DI means degraded images. RI means restored images.

object regions and features are not noted in the training of two processes. Imprecise SR is performed for detection. The detection results are constrained.

According to the above researches, the two tasks are trained just with loss information delivery. This method as multi-task learning (MTL) [9] is not adequate for improving the performance. The points of previous methods are designing more effective networks for the restoration itself. However, just changing the model structure or adding effective modules for SR may not accurately restore detection-interested object features, meanwhile sacrificing the inference speed. Because the guidance from object information is insufficient in SR, SR does not focus on object regions and restores useless features in degraded images. Specifically, problems can be summarized as follows: First, SR lacks the guidance from the spatial information of objects. Due to the scale difference between object regions and background regions, the output score of the discriminator and feature-level restoration of the generator may reflect in background regions; Second, SR lacks the guidance from the detection-interested feature information of objects. Because of intricate details of aerial imagery, extracted features in the discriminator and restored features in the generator may not be related to subsequent detection.

Due to the mentioned problems, restored images (RIs) are generated without guidance from the detection-interested object information. The detection performance is also constrained. In this article, considering OD can provide the needed information for SR, we propose a closed-loop structure to improve the performance via establishing feedback connections from OD to SR. The object information can be delivered to SR based on the structure. Given the mutual reinforcement principle, SR is optimized with the guidance of the object information to enhance restoration, and then restoration-enhanced images are input into OD to improve detection results excellently. The two tasks are optimized iteratively by reducing their own losses. Finally, the overall performance of the structure can be fully exploited.

Therefore, different from the existing methods, the novel learning method is denoted as mutual-feed learning (MFL) as shown in Fig. 2. MFL is proposed to simultaneously train an image restoration task (IRT) and an image interpretation task (IIT). There exists not only the normal connection from IRT to IIT but also the connection from IIT to IRT. These two tasks can feed more valuable information to each other based on a closed-loop structure. The performance of the structure can be further improved with the guidance of information prior. In this article, the connections are established from OD to SR. On one hand, we consider that region of interest (ROI) features from OD contain sufficient spatial and feature information of objects. ROI features can be fed back to teach the discriminator to use the differences in object regions for more effective scores. On the other hand, we argue that features encoded by the backbone of the object detection network are related to objects precisely. There are differences in multiscale features extracted by the backbone between RIs and CIs. The differences can be narrowed to improve feature-level restoration in the generator. Then restoration-enhanced images are input into OD to extract more accurate object features for better results, and more effective guidance can also be provided to SR. In addition, there is no increase in model complexity, and it is necessary for a faster inference speed. In summary, the key contributions are as follows.

1) A closed-loop structure based on an iterative learning strategy is proposed for better detection results in resolution degraded images. It is denoted as MFL. Compared with the existing methods, there is no increase in model complexity.
2) Feedback of region of interest (FROI) module is proposed to implement region-level discrimination in the discriminator using RoI features from OD. It can deliver object spatial and feature information and make the discriminator produce a more effective score.
3) Multiscale object information (MSOI) module is developed to improve feature-level restoration in the generator via using the backbone of the object detection network. It can guide the generator to restore object-related features accurately.

Extensive experiments are carried out on Northwestern Polytechnical University (NWPU) Very High Resolution (VHR)-10 [10], Cars Overhead With Context (COWC) [11], and Fine-grained Object Recognition (FAIR)1M [12] datasets. An outstanding performance can be obtained in degraded images using MFL. The proposed structure has excellent effectiveness and adaptability for models based on GAN and the two-stage detector.

II. Related Works

Since the proposed framework introduces a new learning method for jointly optimizing SR and OD, the related works are described from three aspects: OD, SR, and SR for OD in this section.

A. OD

The existing methods propose many detection models for different applications. The models can be probably categorized into two classes. Single Shot MultiBox Detector (SSD) [13] and You Only Look Once (YOLO) [14] are the one-stage

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detectors, with a faster predicting speed. YOLO [14] directly turns the detection task into the regression task, using the image pixel to predict the category and bounding box. However, results will be worse than those of the two-stage detectors. SSD [13] is proposed to use a fully convolution network and detect objects on different scales. It can get more accurate results.

Region-convolution neural network (RCNN) series detectors are the main representative two-stage detectors. RCNN [15] is the first model of RCNN series detectors. It is proposed by splitting the detection task into the classification task and the positioning task. A region-proposal module is designed to get better results. But this brings a complex multistage training process. Fast RCNN [16] is proposed via improving the speed through an end-to-end training mode with ROI pooling. Faster RCNN [17] is further developed via region proposal network (RPN) instead of selective search for a faster speed. However, these detectors only detect on a one-level feature map. Feature pyramid network (FPN) [18] is proposed to use features of different scales for detection.

These detectors show great ability when images are clear. However, when the coming images are degraded, the results are often unsatisfactory. Studies [19] and [20] have shown a huge gap in detection results between clear and resolution degraded images due to the differences in high-resolution features.

B. SR

The formation of degraded images can be summarized as follows:

$$DI = (CI * k) + n. \tag{1}$$

In this formula, DI represents the degraded images; CI represents the CIs; k represents the degradation kernel; n represents the noise. The main work of SR is to estimate k. However, because the types of k are complicated and hard to estimate in real world, SR faces a great challenge.

k is modeled directly with prior assumptions through early traditional SR algorithms, such as Gaussian blur kernel and motion blur kernel. They use filtering methods to increase the resolution without prior information of CIs. RIs are often not clear enough. Multilayer perceptron makes this task much better. The fully connecting layer extracts features and adaptively estimates k instead of using prior assumptions. However, the learning ability of it is limited.

The nonlinear modeling approach significantly promotes the improvement of restoration quality. Unlike the linear modeling ability of multilayer perceptron, the potent ability of convolution neural network (CNN) for nonlinear modeling is more satisfactory. Super-resolution CNN (SRCNN) [21] is first proposed using CNN for SR. It uses a three-layer network to process RIs obtained by bicubic interpolation [22]. Kim *et al.* [23] develop a more complex network for SR with the residual structure.

With the development of deep learning, GAN [24] shows a more remarkable ability for SR. Super-resolution GAN (SRGNN) [25] is proposed via designing a perceptual loss to make the generator focus on restoring more texture information. Enhanced SRGAN (ESRGAN) [26] is developed by getting rid of the batch norm layer in the generator and designing an Residual-in-Residual Dense Block (RRDB) module to replace the residual block.

C. SR for OD

SR expects to restore more detailed textures in degraded images for more satisfactory detection results. Shermeyer and Etten [19] perform extensive experiments to show that restoring degraded images is helpful for OD, and two tasks are optimized, respectively. Edge-enhanced SRGAN (EESRGAN) [6] is proposed via designing an edge-enhanced GAN for SR. Courtrai *et al.* [27] incorporate methods of CycleGAN [28] and wasserstein GAN (WGAN) [29] into SRGAN [25] for more effective SR. Mu *et al.* [30] develop gradient guidance via generating gradient maps of images to restore structural information. In these researches, the detection loss is backpropagated to fine-tune parameters of the SR network. Haris *et al.* [31] find that only peak signal-to-noise ratio (PSNR) [32] cannot represent the quality of RIs for OD. It proposes a compound loss adding losses of two tasks for a task-driven SR. Joint-super resolution and vehicle detection network (SRVDNet) [7] is proposed via designing a multiscale GAN to restore degraded images. Courtrai *et al.* [27] and Na and Fox [33] propose to perform object-focused SR on object patches obtained by handcrafted ways. Small Object Detection (SOD)-Multitask GAN (MTGAN) [8] is proposed via using an object detection network to generate ROI object patches adaptively for subsequent restoration and detection. Wang *et al.* [34] introduce the effective effect of SR for OD in the remote sensing field. They also propose an SR model based on multifeature fusion and CycleGAN [28] to enhance images. Zhao *et al.* [35] propose to design a residual structure to restore images for subsequent detection. Noh *et al.* [36] propose to use SR to enhance features instead of images. There are also works of literature [37], [38] using SR as an auxiliary method to improve the performance of OD. In these researches, the detection loss and the restoration loss are trained jointly.

However, the two tasks based on an open-loop structure are with a weak connection. The main research direction is to independently pursue the performance of a single task by increasing the model complexity. It may not be effective for overall performance, and further performance improvement is constrained. Therefore, we propose a new learning method called MFL based on a closed-loop structure without increasing complexity for excellent performance.

III. Methods

In this section, we design feedback connections based on the original open-loop structure. Using the novel closed structure for optimization is denoted as MFL. It establishes the close relationship between two tasks to solve the problem. SR can get guidance of the object spatial and feature information for suitable restoration. OD can be performed in restoration-enhanced images for better results. The main contribution is adding multidimensional feedback connections from OD to SR based on the basic feedforward connection. Then, significant object information can be delivered from OD

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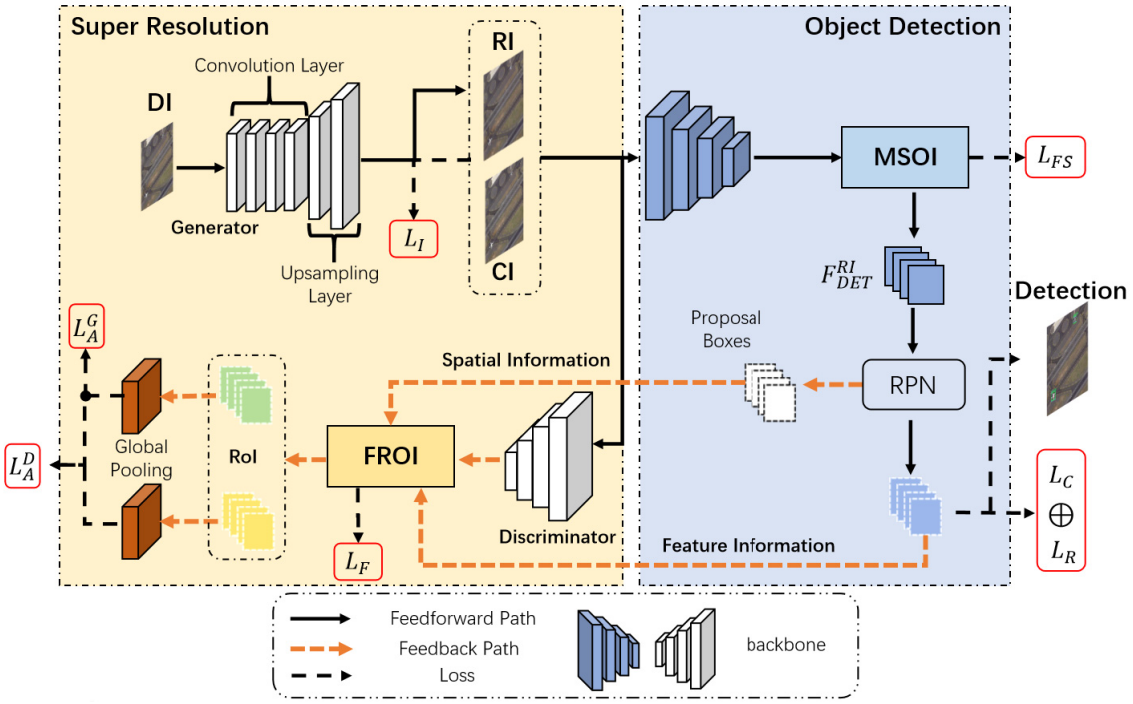


Fig. 3. Whole framework of MFL. There are two parts: the left one is SR, and the right one is OD. Black solid arrows represent the feedforward path, and orange dotted arrows represent the feedback path. In the training process, run on the feedforward path from the generator to the detector, and then continue on the feedback path to the discriminator. In the inference process, only arrows from the generator to the detector are used. The last feature map of RI after MSOI is input into RPN for subsequent operations. When optimizing SR with its loss, the parameters of OD are frozen to avoid the effect. DI, RI and CI, respectively, denote the degraded image, the restored image, and the CI. L_F, L_C, L_R, and L_{FS}, respectively, mean the feedback loss, the classification loss, the regression loss, and the feature similarity loss. L_A^G and L_A^D, respectively, represent the adversarial loss in the generator and the discriminator. F_{DET}^{RI} means extracted features of the backbone in RIs by the detector.

to SR. The following contents will describe the details and optimization functions of MFL.

A. Approach Overview

We propose a novel structure to optimize detection in resolution degraded images better, as presented in Algorithm 1. In this article, to enhance degraded images, SR as IRT is used before OD as IIT. The two tasks are cascaded for better detection. In the existing methods, RIs are generated and input into detection without other operations. The two tasks are trained based on an open-loop structure. The constrained results are obtained due to the little information delivery between the two tasks.

The whole framework of the proposed structure is illustrated in Fig. 3. To fully exploit the performance of the structure, multidimensional feedback connections are applied to feed the information prior from OD to SR. Meanwhile, the enhanced output of SR can provide more valuable features to OD for better results. Therefore, an FROI module and an MSOI module are designed to introduce the object spatial and feature information from OD to SR as the guidance.

Therefore, close connections are established between the two tasks. The optimization of SR can get guidance of object information. SR can pay more attention to detection-interested object regions and restore necessary features for better detection. More excellent detection can provide more effective guidance for SR continuously. Through iterative optimization of the two tasks, the results are improved. As shown in Fig. 4, the closed-loop structure is performed in the training stage.

The proposed structure between the two tasks can further use the information of these two tasks than the original

Algorithm 1 MFL

Require: An initialized SR network, an initialized object detection network, a labeled object detection dataset with degraded images.

Ensure: Restored images, detection results.

1: **repeat**
2: **Step 1: Feedforward connection:**
3: Input degraded images into the generator and obtain restored images;
4: Input restored images into the object detection network;
5: Optimize the object detection network as following (12) and obtain detection results;
6: **Step 2: Feedback connection:**
7: Perform FROI in the discriminator;
8: Optimize the discriminator as following (11);
9: Calculate MSOI as following (6) in the generator;
10: Optimize the generator as following (9);
11: **until** convergence

open-loop structure. The two tasks are iteratively optimized by mutually feeding enhanced results. Therefore, for training two tasks with the cascade relationship, using the closed-loop structure containing the feedforward connection and the feedback connection is denoted as MFL.

B. Feedforward Connection

The details of the feedforward connection are as follows: SR restores degraded images first, and then OD is performed in RIs for detection results. The SR network is based on GAN,

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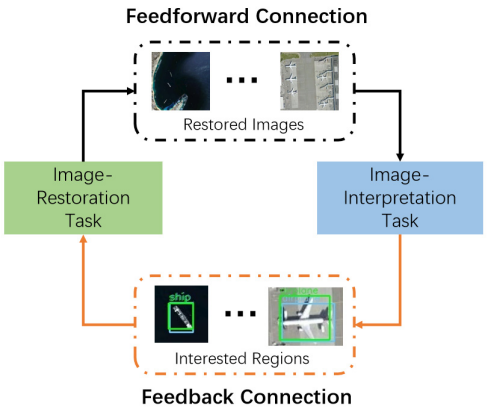


Fig. 4. Feedback connection from IIT to IRT.

and the two-stage detector is selected as the object detection network.

There are two processes in GAN: generation and discrimination. The generation process often uses one convolution layer to deal with the input image first. Second, it uses residual blocks or dense blocks to encode features. Third, there are upsampling blocks for improving the resolution. Finally, it use one convolution layer for changing the feature map channel number to the image channel number. Then RIs are generated. The number of upsampling blocks corresponds to the degradation factor from CIs to degraded images. It often uses the interpolation or pixel shuffle [39] for upsampling blocks. The Visual Geometry Group (VGG) series network is often applied to extract features in the discrimination. The pooling layer usually takes the place of the fully connection layer to reduce computation. A sigmoid function often deals with the final output to obtain a score for the input image.

In the proposed structure, ROI features from detection are needed to guide the discrimination process focus more on detection-interested object regions. Therefore, a typical two-stage detector, Faster RCNN [17], is selected as the object detection network to achieve the feedback operation in subsequent experiments easily.

Only the feedforward connection will not exploit the performance of the whole structure fully. There is adequate information in detection that can have guided SR for effective restoration but is not used. The connection between the two tasks is weak for obtaining better results.

C. Feedback Connection

To feedback important information from OD to SR, three connections are designed, respectively, RPN part to discriminator, ROI part to discriminator, and backbone part to generator. The first and second connection make up FROI module to guide the discriminator to produce more effective scores. The third connection makes up the MSOI module to restore more specific features.

1) FROI: In the original methods, the score in the discriminator is related to the global feature. The discriminator may pay attention to some background regions for producing the score without any guidance. The score may inappropriately reflect the restoration degree for subsequent detection. Therefore, the discriminator is expected to focus more on object regions to produce the score. Meanwhile, the extracted features of object regions should also be related to subsequent detection. Then the score is more meaningful for detection and more accurately represents the restoration degree of object regions. It is considered that the results of RPN part and ROI features in detection contain adequate spatial and feature information of object regions. Therefore, an FROI module is introduced to feed the object region information to the discriminator for a region-level discrimination.

First, the proposal region boxes obtained by RPN are fed back from OD to the discriminator. Second, these boxes are used to crop output features of the backbone in the discriminator, and object region feature patches are obtained. Third, ROI pooling [16] is used to adjust the size of these feature patches for subsequent propagation in the discriminator. Therefore, ROI features representing object region features can be obtained for producing scores.

After CIs and RIs are input into the discriminator, ROI features can be extracted from the pair images. In addition, there are also ROI features extracted in OD. Therefore, we define these three ROI features, respectively, as R_D^{CI}, R_D^{RI}, and R_{DET}^{RI}. Unlike the original methods using features of the backbone containing global information for coarse discrimination, the ROI features containing spatial and feature information of object regions are used for fine discrimination. Therefore, an accurate region-level discrimination for producing scores is achieved.

The discriminative ability for the differences in object regions can be improved via using the obtained ROI features to produce scores. It is more suitable for OD. However, there still exists another problem: using anchors to crop ROI features only provides the spatial information of object regions. The ROI features are still extracted from the discriminator itself aspect. Suppose these features are not related to OD. There are differences in extracted features between the discriminator and OD. In that case, the output score may not be yet effective enough for a detection-driven restoration. The feature information of interest in detection should also be used in the discriminator. Therefore, we propose a feedback loss L_F to enhance the optimization of the discriminator. It contains two distances representing the similarity of ROI features from two tasks. Through using this loss, the discriminator can learn how OD extracts features and what distribution the features fit. Specifically, the features extracted by the discriminator in RIs can be more similar to those extracted by OD; The features extracted by the discriminator in CIs can be more distinguishable from those extracted by the discriminator in RIs. Finally, the discriminator can find differences in detection-interested features in object regions. The output score can represent the restoration quality of object regions more from the object detection aspect. It is better for the detection-driven restoration.

We define the first distance as D_1. It represents the similarity between R_D^{RI} and R_{DET}^{RI} as follows:

$$D_1 = \left\| R_D^{RI} - R_{\mathrm{DET}}^{RI} \right\|_\rho. \tag{2}$$

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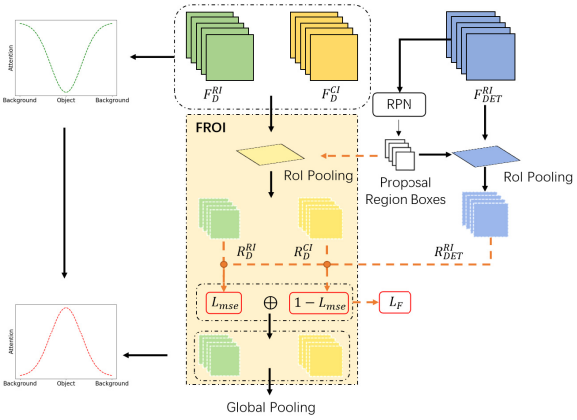


Fig. 5. Details about FROI. The curve represents the attention distribution variation before and after using FROI. F_{DET}^{RI}, F_D^{CI}, and F_D^{RI} denote the features extracted by the backbone of OD in RIs, and the features extracted by the backbone of the discriminator in CIs and RIs, respectively. R_{DET}^{RI}, R_D^{CI}, and R_D^{RI} mean ROI features obtained by OD in RIs, and ROI features obtained by the discriminator in CIs and RIs, respectively.

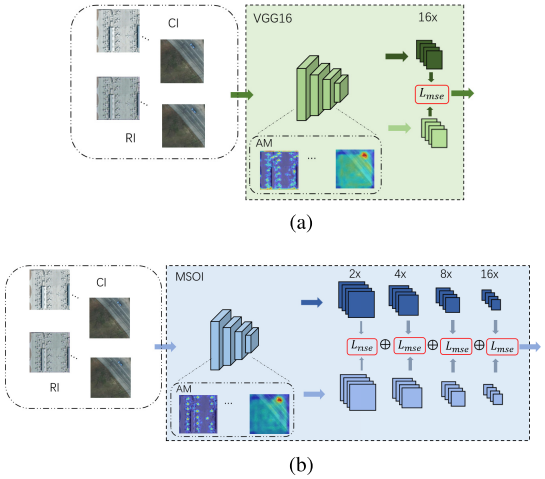


Fig. 6. Structure and AM of (a) original VGG16 and (b) MSOI module. AM means the attention map of the backbone. 2x, 4x, 8x, and 16x represent the reduction factor of the feature map size compared with the input image size. (a) VGG16. (b) MSOI module.

We define the second distance as D_2 representing the similarity between R_D^{CI} and R_D^{RI} as follows:

$$D_2 = \left\| R_D^{CI} - R_D^{RI} \right\|_\rho. \tag{3}$$

For easier optimization, we add the two distances, and define the sum as D for normalization as following:

$$D = D_1 + D_2. \tag{4}$$

Then there comes the proposed loss function as following:

$$L_F = 1 - \frac{D_2}{D} + \frac{D_1}{D} \tag{5}$$

$\|\cdot\|_\rho$ represents the norm for constructing the distance calculation function, and a common norm named L_2 is used in subsequent experiments.

In the optimization process, D_1 is expected to be reduced, and the features of object regions extracted by the discriminator will be more like those extracted by OD; D_2 is expected to be enlarged, and the detection-interested features can be more distinguishable in RIs and CIs. Then the discriminator can extract detection-interested features and simultaneously distinguish pair images based on these features. The generator will restore degraded images based on more effective score of RIs. The RIs will be more like CIs from the object detection aspect. The full implementation of FROI is shown in Fig. 5.

2) MSOI: The feature-level restoration [25] is significant for optimizing the generator. The differences in object features between RIs and CIs need to be narrowed. In this article, detection-interested feature-level restoration should be performed for SR. It is considered that the object detection task can encode features of objects for accurate detection. Therefore, the object detection task can be easily used for object feature restoration in degraded images. Therefore, an MSOI module is developed to guide the generator for restoring detection-interested features.

SRGAN [25] is developed via designing a perceptual loss to restore high-level semantic features in degraded images. A pretrained VGG16 [40] is used to extract features, respectively, in RIs and CIs. Then the two features are expected to be similar for improving the feature similarity between pair images. However, it does not work well for detection-driven SR without specific guidance. There are intricate details and redundant features in aerial imagery. The original perceptual network may restore useless features for subsequent detection. Therefore, MSOI is proposed using the backbone of the object detection network as the perceptual network. The backbone can extract detection-interested features, respectively, in RIs and CIs. Then the feature similarity between pair images will be evaluated by calculating the difference in extracted features. The restoration quality for detection can be quantified and optimized. In addition, multiscale features are necessary to be restored for detecting multiscale objects. Therefore, we select features at four scales from the backbone and calculate the feature similarity in the same scale to obtain a feature similarity loss function L_{FS}. Finally, the loss function is shown as follows:

$$L_{FS} = \sum_{i=1}^{4} \left\| F_i^{CI}, F_i^{RI} \right\|_\rho \tag{6}$$

where F^{CI} means the feature map of CIs. F^{RI} means the feature map of RIs.

The attention map (AM) of the backbone focuses more on object regions than that of the original VGG16 [40] as demonstrated in Fig. 6. It is more valuable and effective for feature restoration from detection perspective. The mean of differences at four scales is not calculated. The features at each scale have its own information and should be affected by other features at different scales. In subsequent experiments, ρ is set as 2 for better results. Through narrowing L_FS, RIs which contain more detection-interested features are generated. Therefore, the detection results can also be improved.

D. Optimization

In this article, to avoid the problem of weight design for the losses of two tasks, we optimize the two networks iteratively.

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To avoid the negative influence for detection, the parameters of the object detection network are frozen when optimizing the SR network. On the contrary, the detection loss is back-propagated to the SR network for fine-tuning. In subsequent experiments, the two networks are trained separately first for subsequent better convergence. Then the two networks are trained jointly based on pretrained parameters.

The details of optimization are described as follows.

1) Optimization of SR: The optimization function of generation contains three components: image loss, adversarial loss, and the proposed feature similarity loss. The image loss represents pixel value differences between CIs and RIs. It can be calculated as given below

$$L_I = \|RI, CI\|_2 \tag{7}$$

where L_I means the image loss. RI means the restored images. CI means clear images.

The adversarial loss is the basic loss in GAN. The generator (G) is expected to generate images which can confuse the discriminator (D), and the discriminator can distinguish real or generated images. The discriminator will, respectively, rate real and generated images. The generator hopes the score of generated images can be close to 1. Therefore, there comes the adversarial loss for the optimization of the generator as follows:

$$L_A^G = 1 - S_{RI} \tag{8}$$

where S_{RI} means the score of RIs. L_A^G means the adversarial loss in the loss of generation.

There comes an overall loss function L_G for the optimization of the generation as following:

$$L_G = \lambda_1 * L_I + \lambda_2 * L_A^G + \lambda_3 * L_{FS}. \tag{9}$$

The hyperparameter setting in subsequent experiments is as follows: $\lambda_1 = 1$, $\lambda_3 = 1$, and $\lambda_2 = 0.1$.

The adversarial loss and the proposed feedback loss make up the optimization of the discrimination. The score of real images is expected to be close to 1, and the score of generated images is expected to be close to 0. The discrimination calculates the adversarial loss as follows:

$$L_A^D = 1 - S_{CI} + S_{RI} \tag{10}$$

where S_{CI} means the score of CIs. L_A^D means the adversarial loss in the loss of discrimination.

There also comes an overall loss function L_D for the optimization of the discrimination as following:

$$L_D = \lambda_4 * L_A^D + \lambda_5 * L_F. \tag{11}$$

The hyperparameter setting in subsequent experiments is as follows: $\lambda_4 = 1$ and $\lambda_5 = 1$.

2) Optimization of OD: OD includes two parts: the classification process and the regression process. They both exist in two stages: RPN and ROI. Therefore, the overall loss function L_{DET} for detection is as follows:

$$L_{\mathrm{DET}} = L_C^{\mathrm{RPN}} + L_R^{\mathrm{RPN}} + L_C^{\mathrm{ROI}} + L_R^{\mathrm{ROI}} \tag{12}$$

TABLE I
PARAMETER SETTING

Stage	Task	Parameter	Value
First	Detection	Epoch	50
		LR	1e-4
	Restoration	Epoch	100
		LR	1e-4
Second	Joint Two Tasks	Epoch	50
		LR	1e-5

where L_C^{RPN} and L_R^{RPN}, respectively, mean the classification and regression loss in RPN; L_C^{ROI} and L_R^{ROI}, respectively, mean the classification and regression loss in ROI.

After optimization of the whole network, RIs are enhanced with the guidance of object information. The detection results can be better in RIs.

IV. EXPERIMENTS

A. Implement Details

The experiments of detection directly in degraded images are first performed to show the effect of resolution degradation. In terms of parameter setting, the training epoch is 50 and the learning rate is 1e-4. Adam [41] is used as the optimizer with the momentum is 0.9 and the weight decay is 0.999. We select the Poly strategy as the way of learning rate reduction. The subsequent experiments are all performed with the PyTorch framework on one 3090 graphics processing unit (GPU) with 24-GB onboard memory.

Then the two tasks are trained jointly. There are two stages for the entire training. In the first stage, two tasks are trained separately for parameter initialization. The object detection network is optimized in RIs generated by bicubic [22]. The SR network is performed under the supervision of CIs. Then the pretrained parameters are saved. In the second stage, the two initialized networks are trained jointly via an iterative optimization strategy.

The parameter setting is demonstrated as given in Table I. The choice of the learning rate reduction strategy and optimizer is the same as the mentioned above. To ensure the greatest possible diversity and complexity of the object categories, extensive experiments are performed over NWPU VHR-10 [10], COWC [11], and FAIR1M [12] datasets to evaluate the proposed structure. The batch size is set to 2 for NWPU VHR-10, 5 for COWC, and 2 for FAIR1M in the training process.

According to the number of object categories in different datasets, average precision (AP) and mean average precision (MAP) are, respectively, used to evaluate the detection results on three datasets. The results are calculated when the standard is $iou = 0.5$. We adopt PSNR and structural similarity (SSIM) [32] for the evaluation of SR results as usual.

B. Dataset Description

To evaluate the proposed structure, extensive experiments are conducted on two coarse-grained datasets: NWPU VHR-10 [10], COWC [11], and one fine-grained dataset: FAIR1M [12].

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TABLE II
RESULTS OF TRAINING IN CIS AND DEGRADED IMAGES ON THE NWPU VHR-10, COWC, AND FAIR1M DATASETS

Image	NWPU VHR-10 MAP	COWC AP	FAIR1M MAP
Clear images	84.84	76.61	44.58
Degraded images (4x)	81.41	71.76	36.78
Degraded images (8x)	70.57	52.04	20.44
Degraded images (16x)	52.43	25.43	0.00

1) NWPU VHR-10: This dataset comprises 800 images. The image resolution ranges from a maximum of 0.08 m to a minimum of 2 m. Objects with different scales in this dataset are categorized into ten classes. The shorter edge of CIs is resized to 600 in subsequent experiments. Then we randomly crop the CIs to a fixed size as 600 × 600 and downsample them to three sizes: 150 × 150, 75 × 75, and 38 × 38. The resolution correspondingly changes from 0.32 to 8 m, from 0.64 to 16 m, and from 1.28 to 32 m in the end.

2) COWC: COWC consists of massive aerial images containing annotated cars. Due to the long shooting distance, cars often belong to small objects. The dataset includes images from six locations: Selwyn in New Zealand, Toronto in Canada, Utah and Columbus in the United States, and Vaihingen and Potsdam in Germany. There are 32 716 unique annotated cars inside, and the resolution is all 0.15 m. The CIs in the dataset have a size of 256 × 256. Sizes of degraded images after downsampling are 64 × 64, 32 × 32, and 16 × 16. At the same time, the image resolution degrades to 0.6, 1.2, and 2.4 m, respectively.

3) FAIR1M: FAIR1M contains 42 796 images which have a ranging resolution from 0.3 to 0.8 m, including 1.02 million objects. All the images are captured by Gaofen satellites or collected from Google Earth. It has a fine-grained category design. All the objects are divided into five classes and 37 subclasses, containing 11 classes for airplanes, nine classes for ships, 10 classes for vehicles, four for courts, and three for roads. The images for training and testing are cropped to 512 × 512. The degraded images are with the size of 64 × 64, 32 × 32, and 16 × 16, and the resolution changes from 1.2 to 3.2 m, from 2.4 to 6.4 m, and from 4.8 to 12.8 m.

In subsequent experiments, 650 images with objects in the NWPU VHR-10 dataset, 1727 images with 6897 cars of Potsdam in the COWC dataset, and 65 663 images cropped from the FAIR1M dataset are used as CIs, and then degraded images are produced by bicubic downsampling CIs. We select 90% of these images to train the whole framework. The remaining 10% of these images are used for test. Before the training process, the operation of flipping and distorting images are used as data augmentation. The proposed structure is applied in different conditions for subsequent comparison and discussion.

C. Training in CIs and Degraded Images

The object detection network: Faster RCNN [17] is first trained in CIs and degraded images with different degradation factors in the NWPU VHR-10, COWC and FAIR1M datasets. To avoid problems brought by too small input size, we first resize degraded images to the size of CIs by bicubic interpolation. The results are shown in Table II.

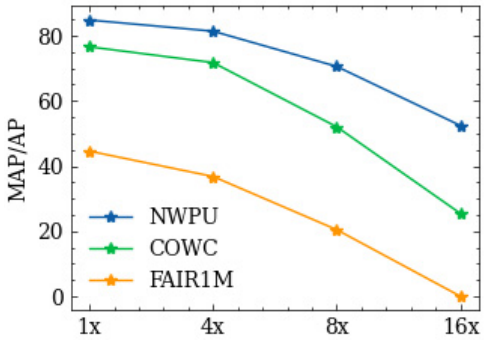


Fig. 7. Results of detection directly in clear and degraded images obtained by different degradation factors; The horizontal axis demonstrates different degradation factors; the vertical axis demonstrates the accuracy.

As presented in Table II, the gap between results in CIs and degraded images is all huge on the NWPU VHR-10, COWC and FAIR1M datasets. Because the resolution of the COWC dataset is higher than that of the NWPU VHR-10 dataset, the detection results can decline slower on the COWC dataset when the degradation factor is 4. Because small objects contain limited information in lower resolution, extracting features becomes more difficult. The results on the COWC dataset decline more than that on the NWPU VHR-10 dataset when the degradation factor is 8 and 16. Moreover, the decline degree of detection results on the FAIR1M dataset is larger than that on both the NWPU VHR-10 and COWC datasets, because more detailed features are needed for distinguishing the fine-grained category.

As shown in Table II, when the resolution keeps degrading, the detection results keep declining. Therefore, performing a detection-driven SR is imperative. These results are applied as the baseline for subsequent experiments. It is hopeful that detection results in RIs can catch up to those in CIs after using the proposed structure. There is also a folding line chart to demonstrate the decline as shown in Fig. 7, making the downtrend more visible.

D. Analysis of Object Feature Difference

There comes a statistical information analysis of the object feature difference. Considering the object features are necessary for detection, the object detection network is used to extract object features for the analysis. Through inputting three inputs: RIs obtained without the proposed structure, RIs obtained with the proposed structure, and CIs into the object detection network, three output features of the last layer of the backbone are obtained, respectively. Then the difference in features between RIs and CIs can be calculated with the mean squared error (MSE) function. Finally, the mean and the variance of the difference are analyzed. It is found that the mean and the variance can be narrowed and more close to 0 after using the proposed structure. The results are shown in Fig. 8.

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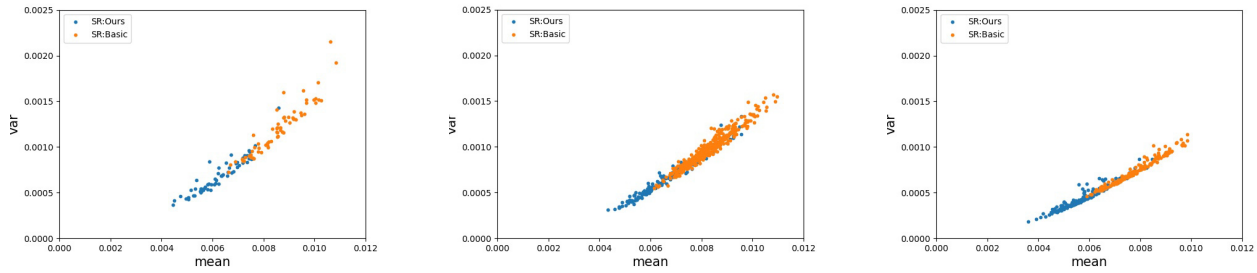


Fig. 8. Statistical information of feature differences between RIs and CIs; the horizontal axis demonstrates the mean; the vertical axis demonstrates the var; Each point demonstrates each image in the testing set. (a) Result on the NWPU VHR-10 dataset. (b) Result on the COWC dataset. (c) Result on the FAIR1M dataset.

TABLE III
EVALUATION OF THE EFFECTIVENESS ON THE NWPU VHR-10 AND COWC DATASETS WITH DEGRADED IMAGES (4X)

Basic Model	FROI	MSOI	NWPU			COWC		
			MAP↑	PSNR↑	SSIM↑	AP↑	PSNR↑	SSIM↑
Bicubic + Faster RCNN	×	×	81.41	26.5	0.743	71.67	27.4	0.721
SRGAN + Faster RCNN	×	×	81.05	25.9	0.708	69.18	27.2	0.714
	✓	×	83.56	26.0	0.712	72.67	27.1	0.700
	×	✓	82.81	27.2	0.767	73.28	28.1	0.736
	✓	✓	**83.93**	**27.3**	**0.769**	**74.17**	**28.1**	**0.737**

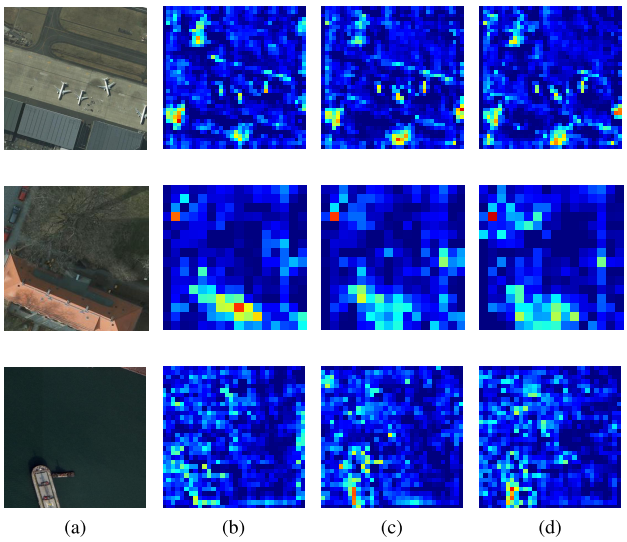


Fig. 9. Feature map visualization of three inputs. (a) CI. (b) Feature map of RI obtained without the proposed structure. (c) Feature map of RI obtained with the proposed structure. (d) Feature map of CI.

Moreover, the features of three inputs are also visualized as shown in Fig. 9. It can be seen that after using the proposed structure, the object detection network can get a larger response in the features of object regions of RIs. Meanwhile, the features of RIs are more similar to that of CIs.

The results indicate after using the proposed work, the object feature difference between RIs and CIs becomes smaller, and better detection is achieved as shown in subsequent experiments.

E. Effectiveness of Proposed Structure

In this section, the effectiveness of the proposed structure—FROI and MSOI module—is proved on the NWPU VHR-10 and COWC datasets. The common model—SRGAN [25]—is used as the SR network. Considering the goal is to evaluate the proposed structure, a faster training speed is preferred. Faster RCNN [17] without FPN is also used as the object detection network. Then ablation experiments are conducted to evaluate whether the proposed structure can obtain better results than that of the basic method.

As demonstrated in Table III, Row 1 demonstrates results using the bicubic method and Faster RCNN. Row 2 demonstrates results with using basic SRGAN and Faster RCNN. The results, respectively, with using FROI or MSOI are presented in Rows 3 and 4, respectively. Finally, using the two modules together can get results as shown in Row 5. These experiments are carried out on the NWPU VHR-10 and COWC datasets.

As can be seen, the methods with using FROI and MSOI can achieve significant results. FROI can help the model achieve surpassed detection results with a 2.51% increase on the NWPU VHR-10 dataset and a 3.49% increase on the COWC dataset. MSOI can help the model achieve surpassed detection results with a 1.76% increase on the NWPU VHR-10 dataset and a 4.10% increase on the COWC dataset. Meanwhile, better PSNR and SSIM can also be obtained after using MSOI. The restoration degree of object regions is also better for human perception. Using the two modules together can achieve surpassed detection results with a 2.88% increase on the NWPU VHR-10 dataset and a 4.99% increase on the COWC dataset. It nearly catches up to the results of training

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Fig. 10. Detection results visualization on images. (a)–(f) Detection results in CIs and RIs obtained by bicubic and basic SRGAN in the NWPU VHR-10 and COWC datasets, respectively. (g)–(l) Detection results in RIs obtained by SRGAN with FROI, SRGAN with MSOI, and SRGAN with FROI and MSOI together in the NWPU VHR-10 and COWC datasets, respectively. (1) Green boxes: correct detection. (2) Cyan boxes: ground truth. (3) Red boxes: wrong detection.

with CIs with only a 0.91% decline of MAP on the NWPU VHR-10 dataset and with only a 2.44% decline of AP on the COWC dataset.

There is detection visualization in different images as shown in Fig. 10. The detection results become more excellent after using the proposed structure. For example, as shown in Fig. 10(a), detection in CIs behaves well. However, there is one wrong detection box in RIs obtained by bicubic [22] in Fig. 10(b) and two wrong detection boxes in RIs obtained by basic SRGAN [25] in Fig. 10(c). After using FROI and MSOI, wrong detection anchors miss, and detection obviously becomes accurate in Fig. 10(g)–(i).

There comes an analysis of the feature AM qualitatively to show that better images with more object information are restored and can provide features for better detection. As presented in Fig. 11, it is apparent that attention becomes less on background regions and more on object regions after using the proposed structure. The corresponding detection results become better. As demonstrated in Fig. 11, the basketball court region cannot be focused on, and the complex background takes up most attention in RIs obtained by bicubic [22] and basic SRGAN [25] in Fig. 11(b) and (c). After using the proposed structure, the attention shifts from the background to the basketball court region in Fig. 11(g)–(i).

To show the outstanding performance, there is a visualization of some detection results on two datasets as demonstrated in Fig. 12

F. Effect of Different CI Resolutions for the Proposed Structure

In this section, the effect of the original CI resolution is studied. There is a comparison of detection results for a single category: car in the NWPU VHR-10 and COWC datasets as Table IV. The proposed structure can make a greater improvement on the COWC dataset, considering images of the COWC dataset are with a higher resolution than those of the NWPU VHR-10 dataset. Higher resolution can apply more effective information for the restoration process based on the proposed structure.

TABLE IV

EVALUATION OF THE EFFECTIVENESS IN THE NWPU VHR-10 AND COWC DATASETS WITH DEGRADED IMAGES(4X)

Category	Ours	NWPU VHR-10	COWC
		MAP	AP
Car	×	70.21	69.18
Car	✓	**71.57**	**74.17**

TABLE V

EVALUATION OF THE EFFECTIVENESS ON THE NWPU VHR-10 AND COWC DATASETS WITH DEGRADED IMAGES (4X)

Dataset	Ours	NWPU VHR-10	COWC
		MAP	AP
ALL	×	81.05	69.18
ALL	✓	**83.93**	**74.17**

There is also a comparison of results for all the categories as shown in Table V. It shows that the improvement is larger than that for car in the NWPU VHR-10 dataset. Considering NWPU VHR-10 contains many various scale objects, and the resolution is lower. Large-scale objects can supply more features than small-scale objects for detection in this situation, and greater improvement can be achieved. The results are demonstrated in Fig. 13 to make the comparison more visible. It is obvious that the slope of the ascending polyline on the COWC dataset is larger than that on the NWPU VHR-10 dataset.

Therefore, there comes a conclusion that the higher the resolution of CIs, the better the results of the proposed

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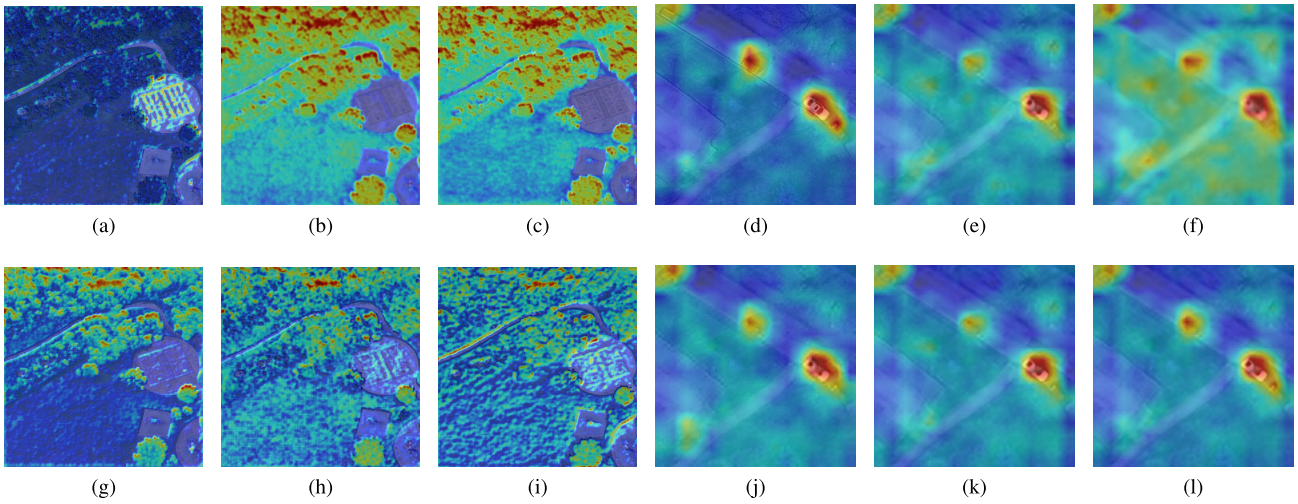


Fig. 11. Feature AM visualization from the object detection network on different inputs. (a)–(f) Detection attention in CIs and RIs obtained by bicubic and basic SRGAN in the NWPU VHR-10 and COWC datasets, respectively. (g)–(l) Detection attention in RIs obtained by SRGAN with FROI, SRGAN with MSOI, and SRGAN with FROI and MSOI together in the NWPU VHR-10 and COWC datasets, respectively.

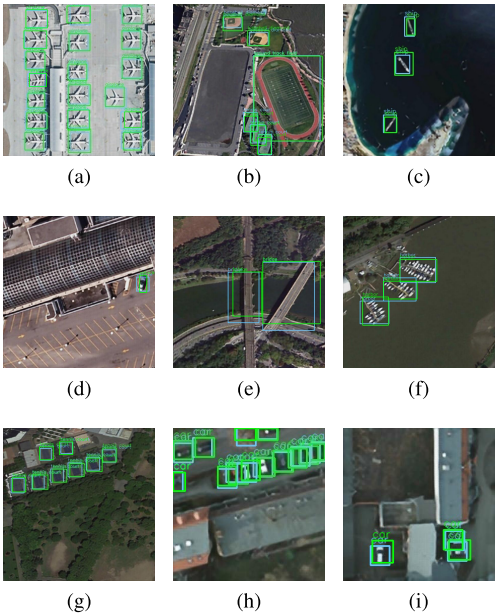


Fig. 12. Detection results' visualization on the NWPU VHR-10 and COWC datasets. (1) Green boxes: correct detection. (2) Cyan boxes: ground truth. (a)–(i) Different selected samples, respectively.

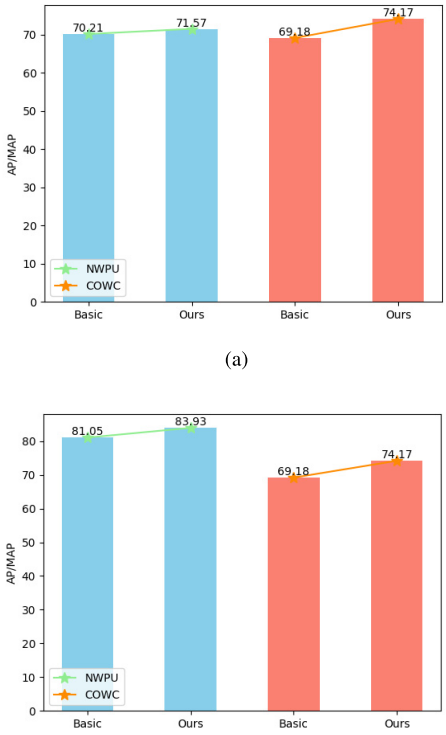


Fig. 13. Results with CIs of different resolution. (a) Results of the category: car in two datasets before and after using the proposed structure. (b) Results of all the categories in two datasets before and after using the proposed structure.

structure. Moreover, small objects are more affected by resolution degradation.

G. Effectiveness of Proposed Structure With Different Degradation Factors

To find the effectiveness of the proposed structure with different degradation factors on three datasets, different degradation factors—8x and 16x—are also selected to perform experiments. We demonstrate the relevant results in Table VI.

As can be seen, the proposed structure also shows effectiveness with 8x and 16x. It can be found that the improvement degree with 8x is better than that with 4x, and the degree of the improvement in 16x is also better than that in 8x. The improvement degree increases as the degradation factor increases. Therefore, there comes a conclusion that the larger the factor of the resolution degradation, the more effective the proposed structure. Fig. 14 is shown to make the visualization of results better.

It can be seen that when the degradation factor is not particularly large as 4, using SRGAN without the proposed structure is not more effective than using bicubic interpolation

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TABLE VI

RESULTS OF TRAINING IN DEGRADED IMAGES OF DIFFERENT RESOLUTION ON THE NWPU VHR-10, COWC, AND FAIR1M DATASETS

Image	Proposed Structure	NWPU VHD-10			COWC			FAIR1M		
		MAP↑	PSNR↑	SSIM↑	AP↑	PSNR↑	SSIM↑	MAP↑	PSNR↑	SSIM↑
Degraded images(8x)	×	71.67	22.0	0.536	58.64	23.5	0.551	25.74	20.1	0.461
Degraded images(8x)	✓	**76.64**	**24.0**	**0.611**	**68.30**	**25.8**	**0.632**	**36.62**	**22.8**	**0.629**
Degraded images(16x)	×	54.57	20.4	0.371	43.39	21.5	0.373	9.24	18.3	0.326
Degraded images(16x)	✓	**65.69**	**21.6**	**0.517**	**55.29**	**23.2**	**0.533**	**24.56**	**20.1**	**0.462**

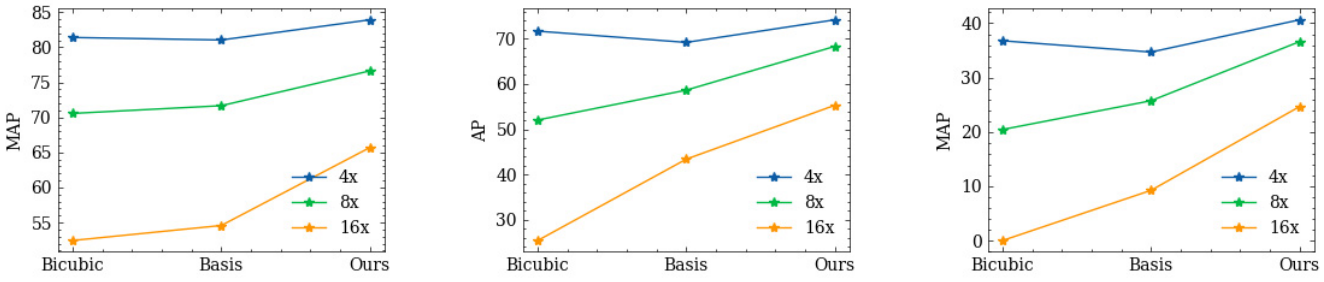


Fig. 14. Detection accuracy variation curve with different degradation factors. (a) NWPU VHR-10. (b) COWC. (c) FAIR1M. "Bicubic" means RIs obtained by bicubic interpolation; "Ours" and "Basis" mean RIs obtained with or without the proposed structure, respectively. 4x, 8x, and 16x represent different degradation factors.

before detection. The bicubic interpolation may retain more raw information in CIs. Using SRGAN without the proposed structure to restore images may damage the information, and subsequent detection may be affected. After using the proposed structure, the object information which is valuable for detection can be introduced into SR, and better results can be obtained. When the degradation factor is set as 8 or 16, the information in CIs is lost severely. The bicubic interpolation cannot restore adequate information for detection. Using SRGAN before detection is more effective, and the performance is much better using the proposed structure.

Optimizing SR without proposed structure is not suitable for detection only.

H. Effectiveness of the Proposed Structure for Fine-Grained Detection

In this section, the ablation experiments are conducted on FAIR1M to validate the effectiveness of the proposed structure in fine-grained detection. The corresponding results achieved on the basis of SRGAN [25] and Faster RCNN [17] are presented in Table VII.

As shown in Table VII, the improvement in fine-grained OD is greater than that in coarse-grained OD. It shows that fine-grained OD has a greater demand for texture details of CIs. Therefore, it is necessary to restore detailed information for distinguishing fine-grained categories.

There comes comparison visualization of detection results and AMs as demonstrated in Fig. 15. It is obvious that after using the proposed structure, the detection results become better, and object regions are also focused more on.

TABLE VII

EVALUATION OF THE EFFECTIVENESS ON THE FAIR1M DATASET WITH DEGRADED IMAGES (4X)

Model	FROI	MSOI	FAIR1M		
			MAP↑	PSNR↑	SSIM↑
Basis	×	×	34.70	23.0	0.625
	✓	×	37.78	22.8	0.628
	×	✓	39.58	24.3	0.704
	✓	✓	**40.64**	**24.6**	**0.716**

TABLE VIII

EVALUATION WITH MORE MODELS IN THE NWPU VHR-10 DATASET

Model	NWPU VHD-10		
	MAP↑	PSNR↑	SSIM↑
ESRGAN + Faster RCNN	82.61	26.3	0.742
ESRGAN + Faster RCNN(ours)	**83.99**	**27.6**	**0.791**
EESRGAN + Faster RCNN	83.87	26.5	0.745
EESRGAN + Faster RCNN(ours)	**84.41**	**27.7**	**0.795**
Joint-srvdnet	82.97	26.1	0.713
Joint-srvdnet(ours)	**83.80**	**27.3**	**0.764**

I. Evaluation of the Generalization of the Proposed Structure

In this section, the generalization of the proposed structure is proven based on more models. There are three models selected: ESRGAN [26], EESRGAN [6], and Joint-srvdnet [7], which are based on GAN for SR. The experiments are

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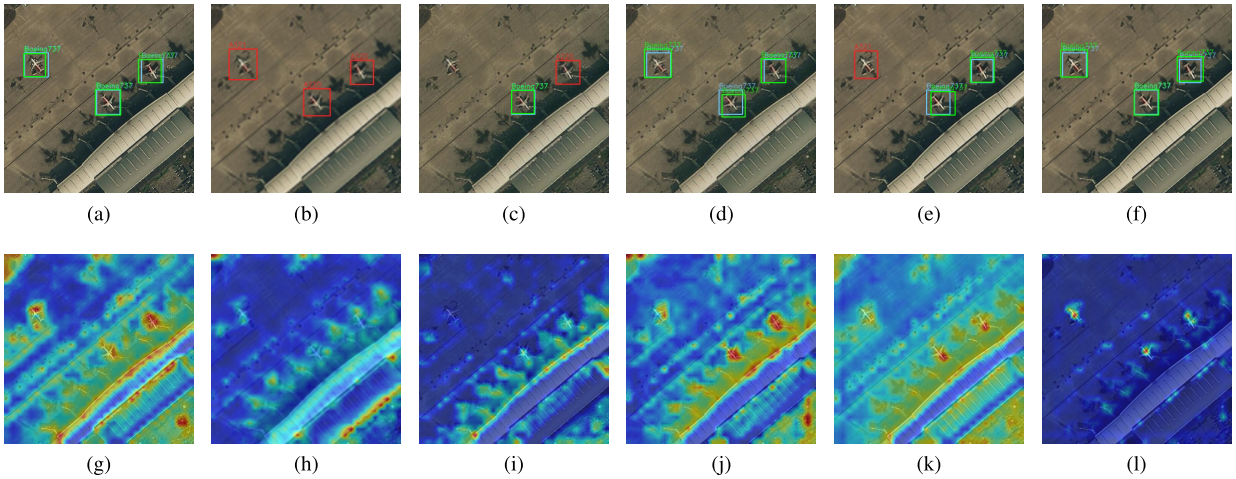


Fig. 15. Detection results and AMs visualization on the FAIR1M dataset. (a)–(f) Detection results in CIs and RIs obtained by bicubic, basis, FROI, MSOI, or FROI and MSOI together, respectively. (g)–(l) Corresponding AM of OD.

TABLE IX
EVALUATION WITH MORE MODELS ON THE COWC DATASET

Model	COWC		
	AP↑	PSNR↑	SSIM↑
ESRGAN + Faster RCNN	71.54	27.5	0.721
ESRGAN + Faster RCNN + ours	**74.21**	**28.7**	**0.765**
EESRGAN + Faster RCNN	72.14	27.7	0.730
EESRGAN + Faster RCNN(ours)	**74.34**	**28.7**	**0.764**
Joint-srvdnet	71.79	27.3	0.714
Joint-srvdnet(ours)	**74.01**	**28.0**	**0.737**

performed on the NWPU VHR-10 [10] and COWC [11] datasets with the degradation factor as 4. Better results are obtained after using the proposed structure compared with the basic models as Tables VIII and IX.

It is obvious that after using the proposed structure, all the three models can get better detection results and restoration degree. It is proven that the proposed structure has excellent generalization.

J. Analysis of Attention Distribution in SR

In this section, the attention distribution in SR before and after using the proposed structure is analyzed on the mentioned three datasets. The variations in the attention distribution are indicated in Fig. 16.

It can be seen that after using the proposed structure, the attention of the generator and the discriminator is more distributed in object regions. The expected objectives are achieved.

K. Comparison and Discussion With Specific Models in the Remote Sensing Field

In this section, these three SR Models: Multi-class Cyclic super-resolution Generative adversarial network with

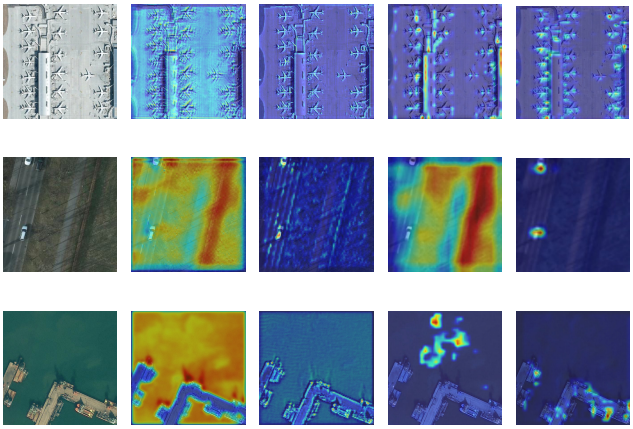


Fig. 16. Attention distribution variation in SR. (a) CIs. (b) AM of the generator without the proposed structure. (c) AM of the generator with the proposed structure. (d) AM of the discriminator without the proposed structure. (e) AM of the discriminator with the proposed structure.

Residual feature aggregation (MCGR) [34], contextual transformation network (CTNet) [42], and edge enhanced GAN (EEGAN) [43], specially proposed in the remote sensing field, are used for comparison and discussion. These experiments are carried over the COWC [11] and FAIR1M [12] datasets. The results are shown in Table X.

It can be seen that through using the proposed structure in specially designed models in the remote sensing field, the detection and restoration results are also improved. Therefore, the effectiveness and generalization of the proposed structure are also verified in the remote sensing field.

L. Analysis of Computation Speed and Model Size

The key idea of MFL is that a novel closed-loop structure is constructed in the training stage, and better parameters of models achieved in the training stage are used for the inference stage. With the proposed structure, no additional module is introduced in the inference stage, and the performance is

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TABLE X
EXPERIMENTS ON THE COWC AND FAIR1M DATASETS

Model	COWC			FAIR1M		
	AP↑	PSNR↑	SSIM↑	MAP↑	PSNR↑	SSIM↑
CTNet + Faster RCNN	73.06	28.5	0.760	38.71	24.2	0.702
CTNet + Faster RCNN (ours)	**75.88**	**28.6**	**0.760**	**42.04**	**24.8**	**0.717**
EEGAN + Faster RCNN	72.04	27.6	0.730	36.51	22.8	0.628
EEGAN + Faster RCNN (ours)	**74.02**	**28.5**	**0.760**	**41.04**	**24.8**	**0.717**
MCGR	73.44	28.0	0.737	37.14	23.0	0.630
MCGR (ours)	**76.09**	**28.5**	**0.755**	**41.76**	**24.6**	**0.716**

TABLE XI
ANALYSIS OF COMPUTATION SPEED AND MODEL SIZE

Super Resolution	Object Detection	Results(AP)		Flops(G)↓	FPS↑	Params(M)↓
		Basis	Ours			
SRGAN	Faster RCNN	69.18	**74.17**	**5.78 + 80.92**	**21.51**	**0.73 + 28.28**
ESRGAN		71.54	**74.21**	17.52 + 80.92	19.94	3.03 + 28.28
EESRGAN		72.14	**74.34**	62.45 + 80.92	15.88	8.59 + 28.28

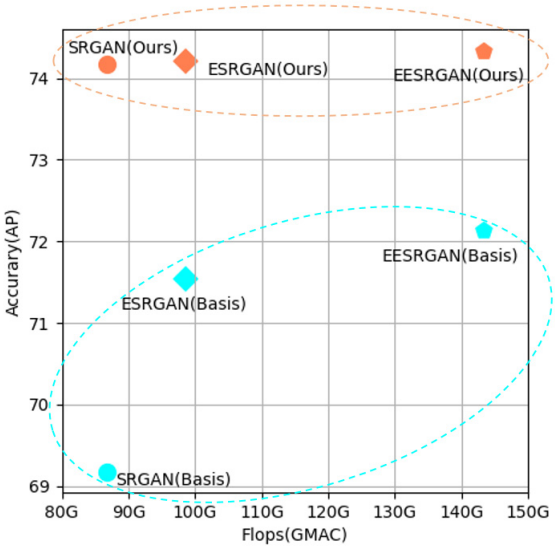


Fig. 17. Relationship between computation speed and detection accuracy. (1) "Basis" means the results without the proposed structure. (2) "Ours" means the results with the proposed structure.

significantly improved without sacrificing speed. Especially when MFL is applied to training when relatively lightweight models are used as the SR part, the performance is close to that of using relatively complex networks. Therefore, the speed is greatly improved in the inference stage for better results.

In this section, to verify the above points, the comparison between SRGAN [25], ESRGAN [26], and EESRGAN [6] in computation speed and model size is presented here. Two indicators—Flops (G) and Params (M)—are used for the evaluation of computation speed and model size, respectively. In addition, the detection results on the COWC dataset are selected for more comprehensive comparison analysis, when the degradation factor is set as 4. The relevant results are shown as Table XI. The part before the plus sign represents SR, and the part after the plus sign represents OD.

As can be seen, with the introduction of MFL, the performance of three various structures can all be improved. More importantly, when the structure consists of relatively lightweight networks as the SR part, the performance can reach that of relatively complex networks. Specifically, SRGAN [25] has only 0.73 M parameters and 5.78 Giga multiply accumulation (GMAC) Flops. It is nearly 4 times and 12 times faster than ESRGAN [26] and EESRGAN [6], respectively. In addition, ESRGAN [26] is nearly four times faster than EESRGAN [6]. With the proposed structure, though the model complexity of EESRGAN [6], ESRGAN [26], and SRGAN [25] progressively becomes lower, the detection results obtained using relatively lightweight models can catch up to that obtained using relatively complex models. Therefore, satisfactory results can be obtained with a faster computation speed.

To make the comparison analysis more obvious, masked with the tag of the SR model, the scatter plot visualization is also shown in Fig. 17. Therefore, this task can be performed on various platforms with the proposed structure for better accuracy according to the actual demand.

V. CONCLUSION

In this article, a novel learning method denoted as MFL is proposed for SR and OD in resolution degraded images. In the existing methods, an open-loop structure constrains the performance due to the little information delivery between the two tasks. Therefore, a closed-loop structure is designed for solving the problem. The detection-interested object information is fed from OD to SR. SR focuses more on object regions for effective restoration with the guidance of object information. Specifically, through designing an FROI module and an MSOI module to enhance the optimization of the discriminator and the generator, respectively, a region-level discrimination of object regions and an object-related feature-level restoration are achieved. The RIs are enhanced for subsequent detection, and better performance is achieved.

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Extensive experiments show excellent effectiveness and generalization of the proposed structure. The improvement of results becomes greater as the resolution degradation factor increases. Furthermore, the proposed structure can perform more effectively, especially in small OD or fine-grained OD. In future work, the interpretability and visualization of the object information will be investigated in depth, and more efficient feedforward connections will be designed.

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